The Induction Principle — problem 8.22

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Consider all possible subsets of the set $\{1, 2, \ldots, N\}$, which do not contain any neighboring elements. Prove that the sum of the squares of the products of all numbers in these subsets is $(N+1)! - 1$. (*Example* : $N = 3$. Then $1^2 + 2^2 + 3^2 + (1 \cdot 3)^2 = 23 = 4! - 1$.)